Faculty of Science, Engineering and Built Environment

SEJ104 Engineering in Society

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

This unit focuses on the principles and practices of human centred design as well as whole system design, within the context of sustainable systems. Design is an essential part of engineering professional practice, and students will explore the process of design ideation, definition and problem-solving, by working on an authentic, real-world problem. The unit allows the students to explore societal, environmental and economic factors while also considering the values and needs of clients and end users including cultural and Indigenous perspectives.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Kaja Antlej
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Other members of the team and how to contact them

Name: Negin Amini, Lecturer
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Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- The strong project and problem-based learning as well as a peer-learning aspect.
- Both the Research and Development (R&D) components based on real-world problems.
- Progressive assessment (feedback) helps students to stay on track and achieve the best results.
- The unit provides a new experience for students and motivates them to come up with innovative and unique designs.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- This trimester, the unit content has been refreshed and narrowed to better suit the future engineering graduates and industry.
- The weighting of the Assessment Tasks has also been amended to better reflect the student workload.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Define problems from a stakeholder and systems perspective and the ways in which they interrelate to develop solution strategies.	GLO1: Discipline-specific knowledge and capabilities
ULO2	Recognize the importance of combining stakeholder expectations and requirements with economic, regulatory and sustainable development objectives when communicating complex ideas and justifying engineering solutions.	GLO2: Communication GLO4: Critical Thinking
ULO3	Demonstrate transferrable employability skills by working collaboratively in a team to implement an authentic real world project that addresses environmental, social and economic aspects of sustainable design.	GLO5: Problem Solving GLO7: Teamwork
ULO4	Demonstrate capacity for integrating engineering principles, ethical, cultural and Indigenous perspectives and understanding of societal impacts when making engineering decisions and developing solutions to address diverse stakeholder needs	GLO8: Global Citizenship

Assessing your achievement of the unit learning outcomes

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Case study review (individual)**
Brief description of assessment task	Students must develop a succinct case study report adhering to a specified format based on a specific area of interest

Detail of student output	This assessment task requires students to prepare a report no longer than 1,000 words. Students will bring in aspects of information researched, and design strategies and techniques used in presenting a case study and why it is a useful solution from both the engineering and social point of view. The design report should include: • Analysis of the scenario • Reflection on the user/society needs • Discussion of design strategies and approaches
Grading and weighting (% total mark for unit)	25% - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Students will identify a need and solution features of interest to a user.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - The ability to critical review and assess examples provides students a insight into design for the engineering profession
How and when you will receive feedback on your work	Students are encouraged to seek regular feedback from their academic mentor. Students will receive feedback on their report within 2 weeks of submission.
When and how to submit your work	Due **Monday 31 March, 2025 8pm (AEDT)** (Week 5) via the Unit site.

Summative assessment task 2

Title	**Career and professional development reflection**
Brief description of assessment task	Students are required to to submit a short reflection report.
Detail of student output	Students are required to attend an Industry Day organised in Week 8 (Friday) and write an individual reflection on how their engagement with this event can inform their career and professional development (500 words). Industry Day will provide an opportunity for the students to actively engage with external guests from different engineering-related organisations who will provide valuable insights into engineering profession. The guests will include representatives from various backgrounds, including First Nations Peoples.
Grading and weighting (% total mark for unit)	10% – marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO4 – Self-awareness is an important skill of a professional engineer serving diverse society.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO8 – Understanding what the role of a professional engineer within a diverse society is essential for a successful career.
How and when you will receive feedback on your work	Students will receive feedback on their reflections within two weeks of submission via the Unit site.
When and how to submit your work	Due **Monday 12 May 2025, 8pm (AEST)** (Week 10) via the Unit site.

Summative assessment task 3

Title	**Visual poster presentation**
Brief description of assessment task	Students are required to develop, submit and present a poster.
Detail of student output	Development of a community information poster that conveys the presented real-world problem and solution via primarily visual means. Oral presentation of the above mentioned poster.
Grading and weighting (% total mark for unit)	30% - marked and graded (15% visual presentation, 15% oral presentation)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO2 – Communicate complex problems and justify solutions to a broad community that considers their expectations and requirements. ULO4 – Identify and develop skills related to engineering professional practice including an awareness of how engineering decision making and practice influences society, including consideration of cultural and Indigenous perspectives.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO2 – Students will need to communicate an engineering solution to a diverse community in a variety of formats. A poster provides an opportunity for pictorial representation of a complex engineering problem and solution. An oral presentation provides an opportunity to communicate relevant and targeted information. GLO4 – A reliable interpretation on your communication must be an outcome. GLO8 – Understanding diverse communities and challenges is a critical skill to all engineers.
How and when you will receive feedback on your work	Students will be provided feedback during the studio sessions to help them improve. Students can also make an appointment to get feedback on their poster.
When and how to submit your work	Due **Monday 19 May 2025 8pm (AEST)** (Week 11) via the Unit site. Oral presentations will be scheduled after the submission **during Studios in Week 11**.

Summative assessment task 4

Title	**Sustainable design project (group)**
Brief description of assessment task	This assessment task requires student teams to ideate, define strategies, document and present strategies and approaches used for addressing the real-world presented problem. Students must contribute to group work by individually locating and collecting information from a variety of sources around topics that are identified during class. They are required to use this information to define the problem from a client and user perspective and present their findings through a project report using a structured format.
Detail of student output	Student groups are required to submit project reports of approximately 3,500 words with appropriate references to source material. The design project should include drawings, sketches, images, photos, and reflections. The design project must address: • Investigation and research conducted • Identification and definition of a problem • Review of a variety of information that addresses the needs of the client and user • Definition of the real-world problem • Design of the group project as a solution to the real-world problem • Intended outcomes • Project management and timelines • Project evaluation plan
Grading and weighting (% total mark for unit)	35% marked and graded. (10% work-in-progress presentation, 25% report submission) *There will be an individualised scaling factor applied based on peer and self-review to reflect your contribution and ability to work well within a team, moderated by the Unit Chair. If this self and peer review is not completed, a penalty of up to 25% of the available marks is deducted.*

This task assesses your achievement of these Unit Learning Outcome(s)	ULO2 – Communicate complex problems and justify solutions to a broad community that considers their expectations and requirements. ULO3 – Students are required to work as effective team members to complete the project, and develop a project management plan, adhering to a standard of professional collaboration through the completion of the project. ULO4 – Students are required to consider the broader societal impact of their solution to the real-world presented problem, as well as showing an awareness of some of the ethical considerations in their practice.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO2 – Students will need to communicate in a variety of formats to convey information, developing a collaborative report in industry standard. GLO4 – Students will need to demonstrate critical thinking to justify their engineering solutions. GLO5 – Students will need to develop a solution to an engineering problem. GLO7 – In industry you will work in teams with people you may or may not know, this task will expose students to this situation. GLO8 – Students will need to consider how engineering decisions and judgements impact society in the context of the presented real-world problem.
How and when you will receive feedback on your work	Students are encouraged to seek regular feedback from their academic mentor and will need to provide evidence of progress in the studio sessions. Students will receive verbal feedback immediately following the work-in-progress presentation in Week 8. Students will receive feedback on their report within 2 weeks of submission. Students will also be provided constructive feedback from their peers regarding their teamwork skills.
When and how to submit your work	Work-in-progress **online or on-campus presentation in Week 8** during Industry Day activities. Report submission **due Monday 26 May 2025, 8pm (AEST)** (Week 12) via the Unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 2 hour professional experience (studio) per week. Students are required to attend and participate in an online industry engagement day for this unit on the scheduled day during trimester intensive activities.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online and collaborative learning including a 1 x 2 hour professional experience (studio) per week. Students are required to attend and participate in an online industry engagement day on the scheduled day during trimester intensive activities as detailed in the unit site.

Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

Learning and assessment activities in this unit for campus students include studio time, designated activities, assessment tasks, readings and study time. For online students the time should be divided between online learning activities (studio time), discussion boards, designated activities, assessment tasks, readings and study time. All students are required to attend and participate in an industry engagement day, either online or optional in-person at either the Waurn Ponds or Burwood campus during the trimester intensive activities as detailed in the unit site. Students may also opt to provide their Assessment 4 work-in-progress presentation online or on-campus during the industry engagement day, further details will be provided on the unit site and in class.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.
Unit learning resources

Your unit learning resources can be accessed from your unit site.
There is no prescribed text. Unit materials are provided via the unit site. This includes unit topic readings and references to further information.
Essential learning resources

This unit follows a project-orientated design-based (PODBL) learning approach. Therefore, no specific articles or books are required for this unit. All resources will be provided during the studios and on the Unit site.
Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.
The recommended resources will be based on the specific project introduced at the beginning of the trimester during the studios and on the Unit site.
Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.
Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025

Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Special learning activities	Assessment due date
1	3 March 2025	Unit and Case Study introduction	Interactive lecture with discussion	
2#	10 March 2025	Design for Society	Interactive lecture with discussion	
3	17 March 2025	Design for Environment	Interactive lecture with discussion	
4	24 March 2025	Case Study Review Writing	Interactive lecture with discussion	
5	31 March 2025	Sustainable Design Project Introduction	Interactive lecture with discussion	**Assessment 1: Case Study Review (Individual)** due Monday 31 March
6	7 April 2025	Teamwork	Interactive lecture with discussion	
7*~	14 April 2025	Indigenous Perspectives	Interactive lecture with discussion	
8	28 April 2025	Industry Day	Interactive lecture with discussion	**Assessment 4: Work-in-progress online or on-campus presentation** during Industry day activities
9	5 May 2025	Project	Interactive lecture with discussion	
10	12 May 2025	Visual Communication	Interactive lecture with discussion	**Assessment 2: Career and professional development reflection** due Monday 12 May
11	19 May 2025	Oral Communication	Poster presentations	**Assessment 3: Visual Poster Presentation** due Monday 19 May **Oral presentations** will be scheduled during studios in Week 11
12	26 May 2025			Study period **Assessment 4: Sustainable Design Project (Group)** due Monday 26 May
13-14^	2 June 2025			End-of-Unit Assessment and exam period

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed